Scientific Density Specification Index

Stage 11E0a–11E8a closeout analysis stack

mdstats

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Specification map

Document	Ownership	Status
field protocols (PDF)	`mdstats.analysis.density_protocols`	implemented
scientific resources (PDF)	`mdstats.analysis.density_resources`; rendering counterpart in `mdstats.plotting.density_resource_policy`	implemented
analysis facade (PDF)	`mdstats.analysis.density_facade` and package exports	implemented
common grid geometry and numerical diagnostics (PDF)	`mdstats.analysis.density_grid_geometry`, `diagnostics`, `stencil_diagnostics`, and `broadening`; plotting compatibility adapters	implemented GR0
common budgeted grid planning (PDF)	`mdstats.analysis.density_planning`; plotting finest-shape compatibility adapter	implemented GR1
plotting grid adaptation (PDF)	`mdstats.plotting.density_visual_policy`; atomic/framework visual compatibility over GR0/GR1	implemented GR2
fixed-kernel scientific grid refinement (PDF)	`mdstats.analysis.density_refinement`; field, basin, and corridor certificates	implemented GR3
scientific grid refinement and plotting reuse (PDF)	implemented GR0–GR3 foundations and planned GR4–GR5 selection/ownership migration	partial

Document	Ownership	Status
periodic species density (PDF)	`mdstats.analysis.density.species`	implemented
attractors and supported basins (PDF)	`mdstats.analysis.density.attractors`	implemented
local mean-force refinement (PDF)	`mdstats.analysis.density.force_refinement`	implemented
provisional temporal assignment (PDF)	`mdstats.analysis.density.temporal_assignment`	implemented
joint evidence validation and structural association (PDF)	`mdstats.analysis.density.evidence_validation`	implemented
species-dependent coordination fingerprints (PDF)	`mdstats.analysis.density.coordination_fingerprints`	implemented
geometry-conditioned site refinement (PDF)	`mdstats.analysis.density.geometry_conditioning`	implemented
final hysteretic segmentation and residence statistics (PDF)	`mdstats.analysis.density.final_segmentation`	implemented
observed transition paths and collective diagnostics (PDF)	`mdstats.analysis.density.transition_paths`	implemented
observed periodic network and transfer validation (PDF)	`mdstats.analysis.density.observed_network`	implemented
Na-LTA NVE-continuation pilot dossier (PDF)	`mdstats.analysis.density.pilot_audit`	implemented
real-trajectory source bootstrap	`mdstats.analysis.density.pilot_execution`	implemented
framework-registered density and attractor pilot	`mdstats.analysis.density.pilot_density_attractors`	implemented
density refinement and attractor lineage	`mdstats.analysis.density.pilot_refinement_lineage`	implemented
structural mapping and temporal-support preparation	`mdstats.analysis.density.pilot_structural_temporal`	implemented
force-density and transition-path readiness	`mdstats.analysis.density.pilot_force_paths`	implemented

Document	Ownership	Status
Stage 11E8a closeout and regression closure	cross-module pilot and density-resource boundary	implemented

The plotting-era atomic-density, framework-density, kernel, sparse-field, planning, and runtime-resource specifications remain normative numerical oracles for kernels that have not yet migrated into analysis-owned modules. Revision 56 completes GR3: common grid geometry and diagnostics, target/finest-feasible planning, exact nested ladders, backend-independent field identities, and fixed-kernel field/basin/corridor certificates are analysis-owned, while signed atomic/framework visual adaptation remains plotting-owned. GR4–GR5 remain planned for cross-fitted numerical-hypothesis freezing and later ownership migration. Stage 11E0a establishes the ownership boundary; Stage 11E1 owns the registered periodic species field; Stage 11E2 owns deterministic supported topology downstream of that field; Stage 11E3 owns PMF-admissible local force refinement without changing the spatial catalog; Stage 11E4 owns provisional temporal evidence without nearest-center filling or final event certification; Stage 11E5 owns orthogonal joint-evidence validation, source-bound structural association, conservative exchangeability checks, and the frozen-versus-refit catalog boundary without final event certification; Stage 11E5a owns exact physical M–O/M–T fingerprints, harmonic diagnostics, occupancy-conditioned mixtures, and conservative structural classification without moving the frozen E5 catalog; Stage 11E5b owns optional one-pass framework-conditioned center regression, translated nested regions, static/dynamic counterfactual memberships, moving-boundary diagnostics, overlap conflicts, and occupancy bounds without publishing final events; Stage 11E6 owns final hysteretic residence and passage segmentation; Stage 11E6b owns exact registered transition paths, periodic translations, compatible path ensembles, and collective-event diagnostics without rates or a kinetic network; Stage 11E7 owns the observed periodic network, structural-versus-observed comparison, identity-preserving compact transfer models, and fail-closed held-out or external transfer outcomes without state merging, refitting, or rate inference. Stage 11E8a-S0 binds one exact raw trajectory to a physical fixed-cell C0 baseline and E0b Na catalog. Stage 11E8a-S1 selects and validates the all-framework center-of-geometry translation gauge, executes a represented-time-preserving E1 density pilot and one E2 attractor realization, and keeps later evidence fail-closed. Stage 11E8a-S2 executes bandwidth lineage, grid refinement, and reference-cell sensitivity without weakening unstable saddle topology. Stage 11E8a-S3 maps the exploratory central attractors onto actual serrated primitive-ring oxygen polygons and transfers the exact coordinate-identical partition to the full represented-time catalog for provisional Stage 11E4 temporal diagnostics. Stage 11E8a-S4 executes the provenance-strict E3 force-refinement boundary and source-bound E6/E6b readiness gate without promoting inadmissible forces or an unresolved spatial hypothesis.

Stage 11E8a pilot dossier and source execution

`pilot_audit.py` owns the source-bound Na-LTA NVE-continuation pilot dossier, bundled real-evidence preflight, explicit blockers, accepted/unresolved fractions, resource accounting, and deterministic human-readable report. It does not replace a missing raw trajectory with legacy summaries.

`pilot_execution.py` owns Stage 11E8a-S0. Given a normalized real trajectory and its raw path, it hashes the source bytes, validates the exact Na-LTA composition, executes the physical fixed-cell C0 baseline, constructs the compact E0b Na position/force sample catalog, and advances the dossier to `blocked_missing_required_evidence`. It does not execute or infer E1–E7 evidence.

`pilot_density_attractors.py` owns Stage 11E8a-S1: framework-gauge selection and sensitivity, deterministic represented-time quadrature, one E1 density realization, one E2 attractor realization, and the corresponding partial dossier evidence. `pilot_refinement_lineage.py` owns Stage 11E8a-S2: signed Cartesian bandwidth lineage, central-bandwidth grid refinement, reference-cell sensitivity, and fail-closed scale/topology certificates.

`pilot_structural_temporal.py` owns Stage 11E8a-S3: packaged persistent-ring replay, actual serrated oxygen-polygon geometry, deterministic attractor-to-ring candidates, exact spatial-partition transfer to the full-weight sample catalog, and provisional Stage 11E4 temporal support. It does not publish final events, paths, or rates.

`pilot_force_paths.py` owns Stage 11E8a-S4: provenance-strict E3 execution and conditional E6/E6b readiness. It records executed blockers rather than inventing PMF or paths.

The closeout specification freezes the engineering status after S4: deterministic LD6 research limits, exact explicit-grid Phase-A bounds, isolated runtime-resource tests, and optional interactive dependency handling. It does not change the scientific dossier or authorize Stage 11E8b kinetics.

Stage 11E8a private common utilities

`_pilot_common.py` owns deterministic serialization, signing, metadata freezing, array accounting, and evidence replacement shared by S0-S4. It is private and does not alter the public pilot schemas. See `pilot_common_spec.{md,pdf}`.